

ADS1115 used with Thermistors

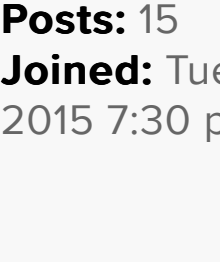
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abishur
Posts: 15
Joined: Tue Jun 16, 2015 7:30 pm

ADS1115 used with Thermistors

by **abishur** »
Tue Jun 16, 2015 9:53 pm

I'm working on a project where I need to measure the temperature of a some water tanks (horse troughs) and record the data for reporting purposes, they are kept at about room temperature (75-ish degrees Fahrenheit).

I'd like to get maximum accuracy possible from my setup. I'm using a 10K thermistor, Amphenol EC95F103W, with +/-0.2% accuracy (though if anyone can tell me where to buy a 0.1% thermistor where I don't have to buy it in packs of 500, I'm all ears) and am looking at the ADS1115 to get 16-bit precision over the 10-bit ADC built into an arduino.

I have two basic questions:

1) If I'm understanding wiring a differential circuit, I would have two wires connect to both sides of the thermistor, the +3v3 and A0 on one side, and GND and A1 on the other. I've seen some diagrams where they further put a 10K resistor on each wire, but that seems like I'd be doing pointless current limiting? Is my understanding of wiring it correct and do I need the resistors?

2) How can I actually, accurately calculate the margin of error at a given temperature? I.e. for the 10K thermistor is lists a 0.00% beta deviation at 25C, and a 0.21% at 30C, and then I need to add in the accuracy of the ADS115. It's less important to me than the first question, but I'd be very interested in learning these things.



adafruit_support_mike
Posts: 69336
Joined: Thu Feb 11, 2010 2:51 pm

Re: ADS1115 used with Thermistors

by **adafruit_support_mike** »
Wed Jun 17, 2015 2:03 am

66 abishur wrote:

I've seen some diagrams where they further put a 10K resistor on each wire, but that seems like I'd be doing pointless current limiting? Is my understanding of wiring it correct and do I need the resistors?

If you connect one end of the thermistor to VCC and the other to GND, the voltage across it will always be VCC, no matter what the temperature is.

If you want to see how the thermistor's resistance has changed, you have to compare it to something. That's what the other resistor is for. It's called a "ratiometric measurement", you don't care about the absolute value that much, you just want to know the ratio of the two resistor values.

The best way to handle thermistor measurements is with a circuit called a Wheatstone Bridge: make one 10k-10k voltage divider from regular resistors and connect the center of that to A0 on the ADS1115. Make another voltage divider with a regular 10k resistor and the thermistor. Connect the center of that to A1. Then set the ADS1115 to take a differential measurement between A0 and A1.

A Wheatstone Bridge is extremely sensitive to small changes in the ratios of the voltage dividers on either side.

66 abishur wrote:

2) How can I actually, accurately calculate the margin of error at a given temperature?

First, make sure your other resistors aren't also changing value with the temperature. Assuming the regular 10k-10k divider is made from resistors from the same production batch, they'll both change roughly the same amount and cancel each other out. The thermistor won't cancel the regular resistor in its divider though, so any change in the regular resistor becomes an error term added to the thermistor's output.

You can solve that by making sure the other resistors don't change much with temperature. The official term for that is "having a low temperature coefficient," and the last two words are almost always shortened to 'tempco'. Ideally, you want a tempco of 0 (the resistance doesn't change at all with temperature), but in practice you'll probably stick with something that has a tempco less than 10ppm/C.

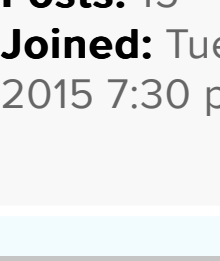
You can get metal film resistors with a tempco of 5ppm/C for about a dollar each: <http://www.mouser.com/Passive-Component...ricing%7C90>

The next step is to get a good voltage reference. A Wheatstone Bridge is also sensitive to changes in the voltage across the resistor dividers, so you don't want that moving around with the temperature too much.

Our LM4040 breakout is about the best thing you'll find without going to one of the really expensive, self-heated references: <https://www.adafruit.com/product/2200> The tempco is typically +/-15ppm/C.

Once you have the major sources of error out of the way, you calibrate the system by exposing it to known temperatures and tracking its output: <https://learn.adafruit.com/calibrating-...-calibrate>

The best possible solution is to fully characterize the circuit by recording its actual output across a range of temperatures that have been measured accurately, taking readings for 1C steps. Do that several times and take the average of the readings at each temperature and you have a lookup table that will convert readings to temperatures with very high accuracy.



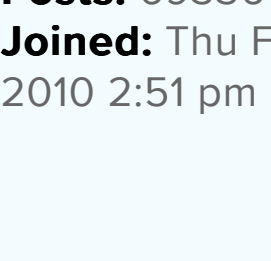
abishur
Posts: 15
Joined: Tue Jun 16, 2015 7:30 pm

Re: ADS1115 used with Thermistors

by **abishur** »
Wed Jun 17, 2015 9:28 am

How would I use the LM4040 with the ADS1115? I thought it didn't have an voltage reference pin and it was handled internally?

Or, now that I'm looking at it again, are you saying I put my 3v3 power on the Vin Gnd to gnd and then use the 2v048 as the + voltage going to the thermistor to provide a constant reliable voltage for measuring?



adafruit_support_mike
Posts: 69336
Joined: Thu Feb 11, 2010 2:51 pm

Re: ADS1115 used with Thermistors

by **adafruit_support_mike** »
Wed Jun 17, 2015 11:03 pm

66 abishur wrote:

are you saying I put my 3v3 power on the Vin Gnd to gnd and then use the 2v048 as the + voltage going to the thermistor to provide a constant reliable voltage for measuring?

Yep. Connect both voltage dividers to the LM4040. Connect the center of one divider to A0, the center of the other divider to A1, and for extra confidence connect the 2.048v line to A3.

Read the difference between A0 and A1 as a differential input, and read A2 as a single-ended input. That will give you an ideal ratiometric measurement because you'll know every voltage that determines the output.



abishur
Posts: 15
Joined: Tue Jun 16, 2015 7:30 pm

Re: ADS1115 used with Thermistors

by **abishur** »
Thu Jun 18, 2015 3:52 pm

Thanks I look forward to trying this all out!

Let me ask one more question, for the Wheatstone Bridge should I keep everything as close together as possible or does it matter? The thermistor will be ~20 feet from the ADS1115 / LM4040. So in one option I could keep as much as it as possible next to the ADS1115 i.e.

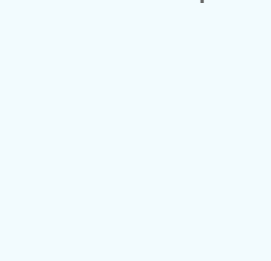
```
CODE: SELECT ALL
LM4040 --- 10K resistor ---- Wire going to A0 -----(20 feet) -----Thermistor -----(20 feet) --- Gnd
LM4040 --- 10K resistor --- wire going to A1 --- 10K resistor --- GND
```

Which would just be 2 wires going outside my enclosure, which would house everything except the thermistor

Or in the second option I could wire everything right next to the thermistor i.e.

```
CODE: SELECT ALL
LM4040 -----(20 feet) -----10K Resistor --- A0 wire --- thermistor --- GND
|----- (20 feet) ----- 10K Resistor --- A1 wire --- thermistor --- GND
```

Which would have four wires going outside my enclosure (Using a common Power and a common Ground). And have the entire Wheatstone bridge outside the enclosure.



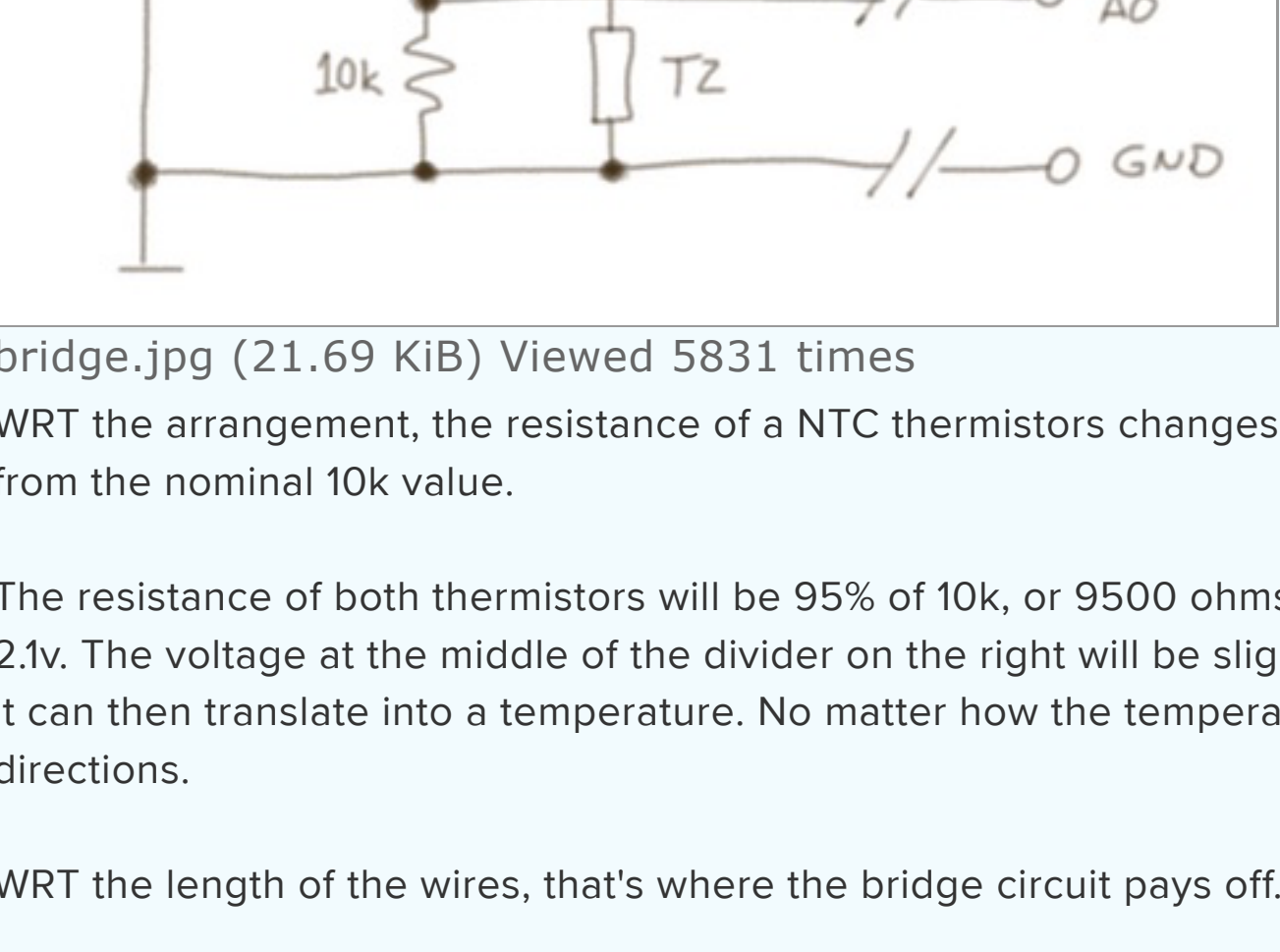
adafruit_support_mike
Posts: 69336
Joined: Thu Feb 11, 2010 2:51 pm

Re: ADS1115 used with Thermistors

by **adafruit_support_mike** »
Fri Jun 19, 2015 2:24 am

Second option., put the LM4040 right next to the bridge.

If you use two thermistors (which is a good idea), arrange things like this:



bridge.jpg (21.69 KiB) Viewed 5831 times

WRT the arrangement, the resistance of a NTC thermistors changes 3% to 6% per degree C, so let's call it 5% and assume we're 1C away from the nominal 10k value.

The resistance of both thermistors will be 95% of 10k, or 9500 ohms. The voltage at the middle of the divider on the left will be slightly over 2.1v. The voltage at the middle of the divider on the right will be slightly less than 2v. That means the ADC will see a difference of 0.1v, which it can then translate into a temperature. No matter how the temperature changes, the voltages on the left and right will move in opposite directions.

WRT the length of the wires, that's where the bridge circuit pays off.

Very little current will flow through the wires for the ADC readings, so the voltage drop across the wires will be small. And since all the wires are about the same length, the voltage drop will be more or less equal for all the wires. That error falls away when you measure the difference between A0 and A1:

$$(v1 + err) - (v2 + err) == (v1 - v2) + (err - err) == v1 - v2$$

So the bridge doesn't care how much resistance you have along the wires, as long as the resistance is about the same in both wires. You can use the voltage at A2 to track any noise being picked up by the cable (EMF from wall power and radio stations), and you can minimize the noise that makes it into the A0-A1 measurement by using twisted-pair cable to carry the signals.

Twisted-pair is arranged so any noise generated by EMF appears equally in both wires, and will drop out of the differential measurement just like the line loss error did. Measuring A2 lets you sanity-check that though. If you see significant variations in that set of readings, you know something went wrong.

BTW - Digikay has the thermistor you mentioned and will let you buy them one at a time: <http://www.digikay.com/product-detail/e...-ND/241331> The package count is 500 if you need to buy a lot of them, but you don't have to buy a whole package (which is good, because that would cost a couple thousand bucks).



abishur
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Re: ADS1115 used with Thermistors

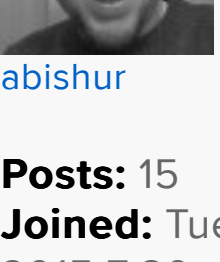
by **abishur** »
Sun Jun 28, 2015 2:04 am

So I've set up everything on a bread board, but I seem to be hitting a snag. First off, I'm powering the ADS1115 with the 3v3 from the Arduino. Will that be causing any problems with the Arduino polling it over I2C at 5v?

Also from the 3v3 of the arduino I've connected to the LM4040 and using the 2v048 output to power the two thermistors, using A0 and A1 as my inputs. My wiring is exactly like we discussed. I've connected 3v3 to vdd, Ground to GND, SCL to the Arduino's SCL, SDA to Arduino's SDA, ADDR to GND, and nothing on ALERT. When using the following Code, I keep getting 0 mV as my reading (I've confirmed that I am getting communications by pinching one of the thermistors and watching the values change). Do y'all have any idea as to what I'm doing wrong?

```
CODE: SELECT ALL
ads1115.begin()
}

void loop()
{
  int16_t Vo; //Reading in mV from ADS1115
  float temp; //Reading as a temperature
  float R; // Vo as a measurement of resistance
  float B = 3892; // B coefficient of Thermistor in Kelvin
  float To = 298.15; // Nominal Temperature in Kelvin
  float Ro = 10000; // Series Resistor (and Nominal Resistance at 25 C)
```



abishur
Posts: 15
Joined: Tue Jun 16, 2015 7:30 pm

Re: ADS1115 used with Thermistors

by **abishur** »
Sun Jun 28, 2015 4:24 pm

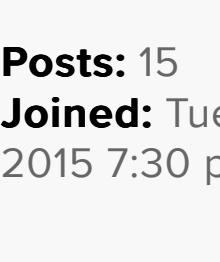
Okay, so yes, having the 5v from the i2c bus was messing with the 3v3 I was using for my voltage input (it was giving me an odd voltage of 4.8ish). Additionally there must have been a problem with the wiring I did because once I pulled everything off and put it back on I had actual readings.... they're wrong mind you, but they're better than 0.

Using a Multimeter I can see that I'm getting 1v662 off my lm4040.

I think the issue I'm having at this point is that my math is wrong. Everywhere I've looked, it's assumed that you're working with 5v on the whole system so Vi and Vo cancel out leaving you with

$$R = \text{SeriesResistor} / (2^{\text{BitOfPrecision}} / \text{ADC_Reading} - 1)$$

But what's the formula to use when your votage source is different than your analog input voltages?



abishur
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Re: ADS1115 used with Thermistors

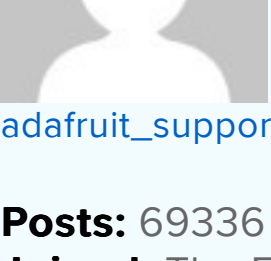
by **abishur** »
Sun Jun 28, 2015 5:39 pm

I think I'm on the right track, I found the following formula

$$Rt = (Rs(Vin - Vout)/(Vout)$$

Where Rt is the Resistance of the thermistor, Rs is a fixed resistor value (10K for me), Vin is the measurement of my analog input, and Vout is the voltage being supplied as my voltage reference.

So for Vout I would still use the 3v3 being supplied to the ADS115. The LM4040 uses a lower voltage, but does so to provide a stable source to the thermistors, not to be used as the voltage reference, right?

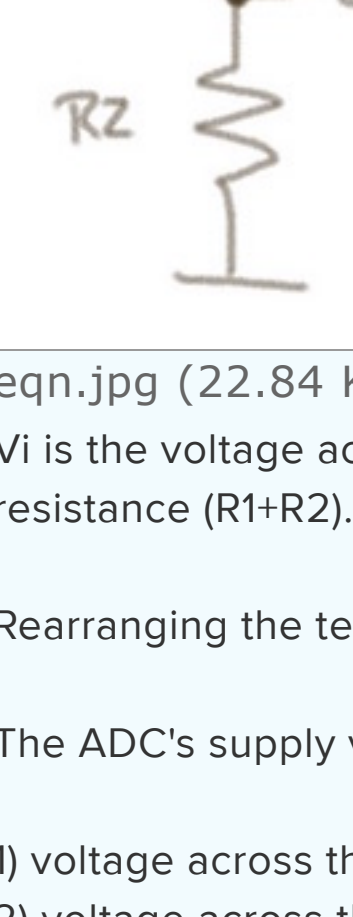


adafruit_support_mike
Posts: 69336
Joined: Thu Feb 11, 2010 2:51 pm

Re: ADS1115 used with Thermistors

by **adafruit_support_mike** »
Sun Jun 28, 2015 10:54 pm

Take it back to the basic voltage divider equation:



$$Vo = \frac{Vi R2}{R1 + R2}$$
$$Vo(R1 + R2) = Vi R2$$
$$Vo R1 = Vi R2 - Vo R2$$
$$R1 = \frac{R2(Vi - Vo)}{Vo}$$

eqn.jpg (22.84 KiB) Viewed 5782 times

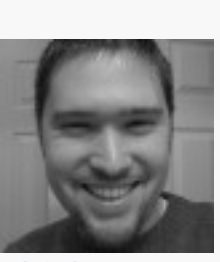
Vi is the voltage across both resistors. Vo is the fraction of Vi proportional to the value of the lower resistor (R2) compared to the total resistance (R1+R2).

Rearranging the terms gives you an equation for the value of R1, assuming you know the values of Vi, Vo, and R2.

The ADC's supply voltage doesn't matter, at least to this equation. If you know any three of the following four values:

1) voltage across the whole divider
2) voltage across the lower resistor
3) value of the upper resistor
4) value of the lower resistor

you can calculate the fourth value.



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Posts: 15
Joined: Tue Jun 16, 2015 7:30 pm

Re: ADS1115 used with Thermistors

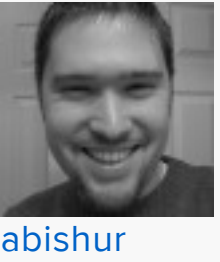
by **abishur** »
Sun Jun 28, 2015 11:43 pm

I know this should be making sense, but I feel like I'm missing something (sorry). So I connect the ADS1115, I'm using the LM4040 to give it a steady 2.048v. Let's look at just one of the voltage dividers at a time to make sure I'm doing it right like you listed. When the ADS115 reads the input of that voltage divider then it's going to return a number between 0 and 65535 (2^16 - 1). Since I'm doing a gain of two the Arduino example sketch shows that I need to multiply that by 0.0625 to have an accurate number.

So if I take the number the ADC gives me and divide it by 65535 then I will have a percentage which I can multiply by Vin to tell me what my Vo is?

Edit: Or to put it another way, my Vo should just be the differential reading off the ADS * 0.0625.

So I need to translate my 2.048 into millivolts and my equation to find R would be R=(2048-Vo)/Vo



abishur
Posts: 15
Joined: Tue Jun 16, 2015 7:30 pm

Re: ADS1115 used with Thermistors

by **abishur** »
Mon Jun 29, 2015 9:25 pm

Or rather, let me change focus here. Maybe the issue is I'm not understanding how the ADC is actually working so let me try asking three other questions and we can work back to the main issue.

I steady the ADS1115 from the 3v3 rail. Also being powered off the 3v3 rail is an LM4040 shunt voltage reference. I give it 3v3, it gives a steady(er) 2v048. Now I would think that this means that I would tell the ADS1115 to use a GAIN 2 on the analog readings, but I see in the data sheet that I should never allow my Analog in value to be greater than my supply voltage +0.3v or 2v348.

So question 1: To take my small range using the 2v048 and make use of the GAIN_TWO settings do I need to supply the ADS1115 with 5v instead?

Let's assume the answer is yes. So now my VDD is 5v, it still goes to the LM4040 and gives me approximately 2v048 which goes off to my Wheatstone Bridge

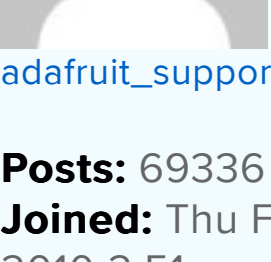
The schematic looks something like this

```
CODE: SELECT ALL
|-----10K Resistor-----Thermistor-----|
+5v ----- LM4040 (+2v048) ----- Thermistor ----- 10K resistor ----- GND
|-----ADS1115-----A0-----A1
```

I take my differential reading off A0 and A1 and I'm given a value somewhere in the range of 0 - 65535

Question 2: Zero is definitely equal to zero volts (my ground), but what is the 65535 value? Does it equal the 2v048 I'm supply to the voltage dividers or is it the 5v I'm using to supply the ADS1115?

Question 3: How does the gain factor into all of this? Do I need to divide or multiply the Analog input reading by something to properly interpret the results? For the tutorial for using thermistors I see it talks about multiplying it by 0.0625 to get the reading in millivolts, but it never explains where it got that number from or what assumptions it is making about my circuit for me to be accurate.



adafruit_support_mike
Posts: 69336
Joined: Thu Feb 11, 2010 2:51 pm

Re: ADS1115 used with Thermistors

by **adafruit_support_mike** »
Tue Jun 30, 2015 12:38 am

The answer to question #2 will probably give you the biggest aha!

When you take a differential reading, GND isn't zero.

A differential reading tells you the difference between the two inputs: A0-A1. An output of zero means literally A0-A1=0.

That means you can't associate any specific voltage on a single input with 'zero'. If the voltage divider has 2.048v across it, an output of '0' means A0-A1=0 for any value of A0 from 0v-from-GND to 2.048v-from-GND.

The output is a signed integer, so an output of 65535 doesn't exist. The range is -32767 to 32767. The bit pattern 0xffff maps to the signed integer -1. The most positive output you can get maps to the bit pattern 0x7fff (32767), and the most negative output you can get maps to the bit pattern 0x0001 (-32767). The advantage of that mapping is that 0x7fff-0x0001=0., adding any number to its negative gives you zero.

So., having established that the differential signal tells you A0-A1, you select a gain that tells you the largest expected difference between A0 and A1.

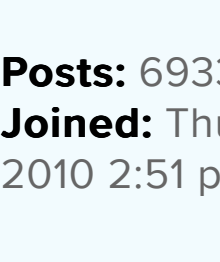
If A0 can be 2.048v-from-GND while A1 is 0v-from-GND (or vice versa), you need to choose an input range that allows A0 and A1 to be 2.048v apart.

If A0 and A1 stay within 0.5v of each other, you can choose a gain that only allows that +/-0.5v of difference.

Once you've selected your gain, you divide it by 2^15 to get your resolution. If your range is 2.048v, the smallest voltage difference the ADC can measure is 2.048/32768==62.5uV. If your range is 0.5v, the smallest voltage difference the ADC can measure is 0.5/32768==15.2uV.

Using the smallest possible gain that can measure the largest possible input swing gives you the best possible output resolution.

Assuming you've chosen the input range of 2.048v, the output value 0 means A0-A1==0 as discussed above. The output -32767 means A0 is 2.048v more negative than A1 (Nv-(Nv+2.048v)==-2.048v). The output 32767 means that A0 is 2.048v more positive than A1 ((Nv+2.048v)-Nv=2.048v).



abishur
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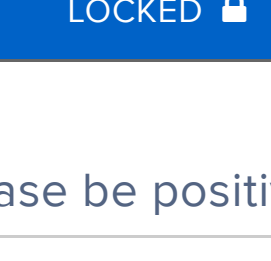
Re: ADS1115 used with Thermistors

by **abishur** »
Tue Jun 30, 2015 12:57 am

Well that was certainly some hands down anal explanations and one more question should give me everything I need to know.

If I was to make use of three of the analog inputs with A0 and A1 being my differential inputs and A2 being a connection to the 2.048 coming off the LM4040 as an absolute measurement of my voltage, how would that affect my gain settings?

Would it be as simple as setting gain to 2 when I read from A2 and then setting it to 8 for my differential readings?



adafruit_support_mike
Posts: 69336
Joined: Thu Feb 11, 2010 2:51 pm

Re: ADS1115 used with Thermistors

by **adafruit_support_mike** »
Wed Jul 01, 2015 6:29 am

If you use a gain of 2 for the 2.048v measurement, you'll only get meaningful output if the input is low. Setting the gain to 1 for a range of 4.096v would give you readings above and below 2.048v, but you'd lose one bit of resolution.

Since you expect the 2.048v reference to be stable, you could cheat it down about 1% with a 100-10k voltage divider. Then you can use the gain of 2 and multiply the reading by 1.01 to take you back up to the correct value.

A gain of 8 is a decent starting point for the differential reading. Collect some data and see if you want to adjust it up or down.